

Supplementary Material for: Federated O-RAN Slice SLA Prediction: A Cross-Architecture Empirical Benchmark on Colosseum/ColO-RAN

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Cross-architecture federated-learning benchmark on Colosseum/ColO-RAN: per-section appendix material moved out of the main paper for length compliance with IEEE journal main-body conventions. Each section heading carries the originating main-paper section in parentheses.

APPENDIX A

MECHANISM ANALYSIS DETAILS (EXTENDS § 7.1)

A. Applicability boundary (extends § 7.1)

The apparent inversion is a *sequence-integrity artifact* of row-level client partitioning (§ 7.1, run-level controls), not a dataset-structural property; its scope is therefore a property of *benchmark construction* rather than of RAN telemetry. It replicates whenever a federated benchmark (a) pools a time series and synthesises client heterogeneity by row-level partitioning (Dirichlet-over-samples or uniform random assignment — the de-facto FL non-IID default), and (b) constructs per-client sliding windows *after* partitioning, so the row split scatters each entity’s consecutive timesteps and corrupts the windows. Deployments and benchmarks that partition by natural entity (each gNB / run kept intact) never incur it and behave classically — the natural partition recovers full AUC because it preserves sequences, not because RAN telemetry has special cell-conditional structure. We conjecture other federated time-series benchmarks that partition by row exhibit the same spurious inversion; auditing them with the run-level controls and the window-contiguity check of § 7.1 is open future work.

B. Hardware drift caveat for the § 7.1 *random_split* ablation (extends § 7.1)

The § 7.1.1 *random_split* ablation cells were run on 4× Tesla V100-SXM2-32GB (sm_70, driver 535.161, CUDA 12.1) while the Phase 5 baseline (natural-by-BS and Dirichlet partitions) was run on RTX 4080 (sm_89). The two GPU generations have different bf16 implementations (V100 emulates bf16 via fp16 tensor cores with possible fp32 fallback; Lovelace has native bf16 tensor cores), and `cudnn_deterministic=True` pins kernel choice within a hardware target but does not impose bit-equivalence across hardware. We bound the resulting AUC drift by comparing

V100 *random_split* AUC to 4080 Dirichlet $\alpha_{\text{dir}} = 5.00$ AUC (both are partition strategies that destroy bs grouping, so they should yield equivalent AUC under hardware-independent training): the residual delta is at most 0.0072 (LSTM), 0.0043 (Mamba), 0.0021 (Spiking-SSM), all within the corresponding seed σ . Hardware drift is therefore bounded above by ~ 0.007 AUC, more than an order of magnitude smaller than the ~ 0.18 mechanism signal (ratio $\approx 25\times$). Strict elimination of hardware confound would require re-running Phase 5’s IID column on V100 (≈ 10 GPU-hours on the 4× cluster); we did not undertake this because the bounded-drift argument is sufficient to interpret § 7.1.1’s ablation conclusion.

APPENDIX B

REPRODUCIBILITY INFRASTRUCTURE DETAIL (EXTENDS § 5)

A. Test infrastructure (extends § 5.2)

The release ships **277 passing tests** across four suites that protect numerical and pipeline-level invariants:

- `tests/test_v7_*.py` — 202 unit + integration tests for `fl_v7` trainer (model-build, partition correctness, scaler determinism, FedDyn canonical-vs-Option-II contract).
- `tests/test_aggregate_v7_results.py` — 22 tests for the aggregator (paired-bootstrap, BLAKE2b decorrelation, JSON round-trip, schema-mismatch detection, 1260-cell scaling).
- `tests/test_paper_draft_invariants.py` — 37 paper-text invariants (forbidden hallucinated facts; required corrected facts; license consistency; hardware specs).
- `tests/test_paper_claims_sources.py` — 16 paper-vs-data claims tests that re-read `artifacts/step1_factfinding.json`, `step2_mechanism_search.json`, and `aggregated_phase5.json` and assert paper-reported numbers match the underlying measurements within rounding tolerance.

The latter two suites (53 tests total) form a paper-correctness developer pre-commit gate: every `PAPER_DRAFT.md` edit is verified by re-running them, preventing quick-claim regressions.

B. Statistical pipeline (extends §5.3)

`scripts/aggregate_v7_results.py` consumes the `per-cell_summary.json` files and produces `aggregated_phase5.json` (90 group means + 270 paired-bootstrap distributions: 180 algorithm-pairs + 90 architecture-pairs) plus a paper-grade Markdown table. Each pairwise comparison’s bootstrap RNG seed is derived from a BLAKE2b hash of the pair identifier added to a base seed (2026 for algorithm-pairs, 2027 for arch-pairs), guaranteeing independent bootstrap streams across all 270 distributions and supporting joint coverage claims (§4.5).

C. Croissant dataset metadata (extends §5.4)

The Colosseum/CoLO-RAN public release is upstream-licensed; our preprocessed `coloran_raw_unified.parquet` ($\approx 18\text{M}$ rows) and the partition specifications are documented in a Croissant 1.0 metadata file (`croissant.json` at the repository root, validated against the official `mlcroissant` validator) shipped with the release archive. The metadata block describes the (`bs_id`, `slice_id`, `sched`, `tr`) keying, the 17 continuous features (§3.1), the OOD train/val/test split (§3.3), and the `add_classification_target` derivation rule (§3.2).

D. Reproducible execution environment (extends §5.5)

The release archive bundles:

- `Dockerfile.repro` — pinned to the exact CUDA / PyTorch / `nvidia-ml-py` versions used in Phase 5 (CUDA 12.8, PyTorch 2.10, `nvidia-ml-py` 12.x);
- `requirements.lock` — SHA256-pinned Python dependencies (74 packages);
- `repro.sh` — one-line entry point that loads the `parquet`, runs a 1-cell smoke (LSTM, FedAvg, IID, `seed=42`, 5 rounds $\times$ 20 max-steps, $\sim 30\text{s}$ on RTX 4080), and verifies bit-equivalent test AUC against a stored reference value within 10^{-6} .

E. Demo notebook (extends §5.6)

A Jupyter notebook (`notebooks/colosseum_oran_federated_slicing_demo.ipynb`) walks through (a) loading aggregator JSON, (b) reproducing Figures 1–3, (c) running a downscaled smoke version of the §7.1.1 `random_split` ablation on a single GPU (the full ablation runs on the $4\times$ V100 setup documented in App. A–B), and (d) regenerating §6 paired-bootstrap CI_{95} from raw cells. The notebook is the recommended onboarding entry point for downstream researchers extending the benchmark.

A note on internal references in C.1–C.6: “Stage 1” refers to the predecessor centralized 3-architecture benchmark on CoLO-RAN (LSTM, Mamba, Spiking-SSM) whose pre-registered hardware budget, precision policy, and gradient-step count are inherited by Phase 5 (the federated extension reported in this paper). “ADR-001 D-N” denotes the N -th preregistered design decision in the project’s Architecture Decision Record. Both artefacts are documented in the release archive for downstream reproduction.

A. L9 full discussion: `pos_weight_split=train` choice (extends §8 L9)

We compute the BCE positive-class weight from the train partition’s positive rate (per ADR-001 D-12), in line with standard practice that prevents test-set positive-rate from leaking into the loss. Two alternative computations (val-derived `pos_weight`; held-out-fold `pos_weight`) would also be valid and might shift absolute AUC numbers slightly, but the BCE loss reweighting is identical (globally pooled) across all clients regardless of which split the rate is computed from, so the *direction* of all our findings is invariant to this choice. (The inverted- α_{dir} itself is a sequence-integrity artifact, §7.1; it does not depend on the `pos_weight` split either.)

B. L10 full discussion: `bf16` mixed-precision training versus `fp32` (extends §8 L10)

All Phase 5 cells use `bf16` mixed precision per the Stage 1 inheritance contract for memory and throughput on RTX 4080. The numerical-precision gap between `bf16` and `fp32` is bounded for our model architectures by the lack of overflow-prone operations (no exponential losses, no large-fan-in matmul without LayerNorm); we do not report a `fp32` verification cell and treat `bf16` vs `fp32` cross-check as a reproducibility-supplementary item for the camera-ready archive. Readers reproducing on hardware without native `bf16` (Pascal `sm_60` and earlier) should expect $O(0.001)$ AUC drift.

C. L11 full discussion: `per-client` compute budget split (extends §8 L11)

Our (round, `max_steps`) configuration matches Stage 1’s audit-corrected 25 000 total-gradient-steps budget. An alternative split (e.g., 50 rounds $\times$ 100 max-steps, 200 rounds $\times$ 25 max-steps) would shift the round-vs-step ratio and potentially affect server-side aggregation count by 2–4 $\times$; we did not sweep this dimension, treating round count as a fixed reproducibility anchor. Whether the FedAdam-saturates-headroom finding (§6.3) survives at $4\times$ more rounds with proportionally fewer steps per round is open.

D. L12 full discussion: SLA-threshold sensitivity (extends §8 L12)

The 10% BLER threshold is the canonical CoLO-RAN SLA gate [1]; the empirical positive rate at this threshold on the train partition is 30.9% (§7.1 setup-level facts). Since the inverted- α_{dir} is a sequence-integrity artifact (§7.1) — per-client window corruption, independent of the label threshold

APPENDIX C

EXTENDED LIMITATIONS AND THREATS TO VALIDITY

(EXTENDS §8)

Each subsection below holds the full prose of an §8 limitation whose main-paper bullet was condensed to a 1–2 sentence headline. The §1 contribution(s) each item threatens are stated in the main-paper §8 bullet headers.

— we expect the corrected (run-level) result to be *threshold-invariant*: run-level partitions should match natural-by-BS at every threshold, while the row-level deficit should persist (it is windowing, not labels). A 5% / 15% / 20% sweep would confirm this threshold-invariance of the corrected finding; we did not run it. (The earlier prediction that “inversion strengthens as the threshold tightens” is withdrawn — it presumed a structural mechanism that the run-level controls refute.)

E. L13 full discussion: bootstrap CI percentile vs BCa (extends § 8 L13)

§ 4.5 explains the choice: at our $n = 10$ paired seeds with approximately symmetric per-seed delta distributions, the bias-correction term in the BCa interval is bounded above by $O(1/\sqrt{n}) \approx 0.32$ standard errors; this magnitude is smaller than our seed σ on every reported finding except § 6.4’s Mamba×SCAFFOLD interaction, where σ inflation by $23\times$ makes the CI choice less consequential than the Wilcoxon p -value. Readers who prefer BCa should treat our CI_{95} as a slightly tighter version of the BCa interval; the qualitative conclusions are unchanged.

F. L14 full discussion: FedAdam hyperparameter sensitivity (extends § 8 L14)

$\beta_1 = 0.9$, $\beta_2 = 0.99$, $\eta_{\text{server}} = 0.01$ follow the FedAdam paper’s preregistered values [2]; $\beta_2 = 0.99$ differs from canonical Adam’s 0.999. We did not run a $(\beta_1, \beta_2, \eta_{\text{server}})$ sensitivity sweep — the FedAdam-vs-FedAvg deltas reported in § 6.3 are anchored to paper-canonical hyperparameter choices rather than to a per-task hyperparameter optimum. Whether the FedAdam-saturates-headroom finding survives at alternative β_2 (e.g., 0.999) is open; we expect modest sensitivity (≤ 0.005 AUC drift) given Adam’s relative robustness to β_2 variation in offline ML literature, but this is anticipated, not measured.

APPENDIX D

FUTURE WORK DIRECTIONS (EXTENDS § 9.1)

Several open questions remain. **First**, the run-level controls (§ 7.1) resolve the inverted- α_{dir} as a sequence-integrity artifact, but a small (~ 0.002 AUC) residual FL-heterogeneity cost remains for run-level Dirichlet at the most-skewed α_{dir} ; cleanly isolating it from per-round-participation confounds (the `sub_per_bs=2 / 5/14`-sampling-fraction design of the per-BS ablation) — e.g. an isolation re-run bumping `clients_per_round` to match Phase 5’s $5/7$ fraction — is minor open work, following the non-IID-impact methodology of Jimenez-Gutierrez et al. [3] and ProFed [4]. **Second**, our 10% BLER threshold should be perturbed to $\{5, 15, 20\}\%$ to confirm operational robustness, in line with recent rare-event metric-stability work [5]. **Third**, scaling beyond $N = 7$ base stations to the larger- N 6G slicing and edge-FL regimes studied by Chiarani et al. [6] and Sezgin et al. [7] would test whether the corrected finding — that FL is robust to client heterogeneity once sequences are kept intact — continues to hold at large N . **Fourth**, the FedAdam headroom-saturation finding may be soft: variance-aware sharpness methods such

as FedSCAM [8] and synthetic-data-guided FedSynSAM [9] introduce per-client perturbation modulation that could extract additional accuracy. **Fifth**, integrating differential privacy via over-the-air channel noise [10] and exploring neuromorphic edge deployment — both sparse-RNN acceleration on Loihi-class accelerators [11] and spiking-NN low-power edge sensing [12] — would close the loop toward production-deployable privacy-preserving energy-efficient O-RAN slice prediction.

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